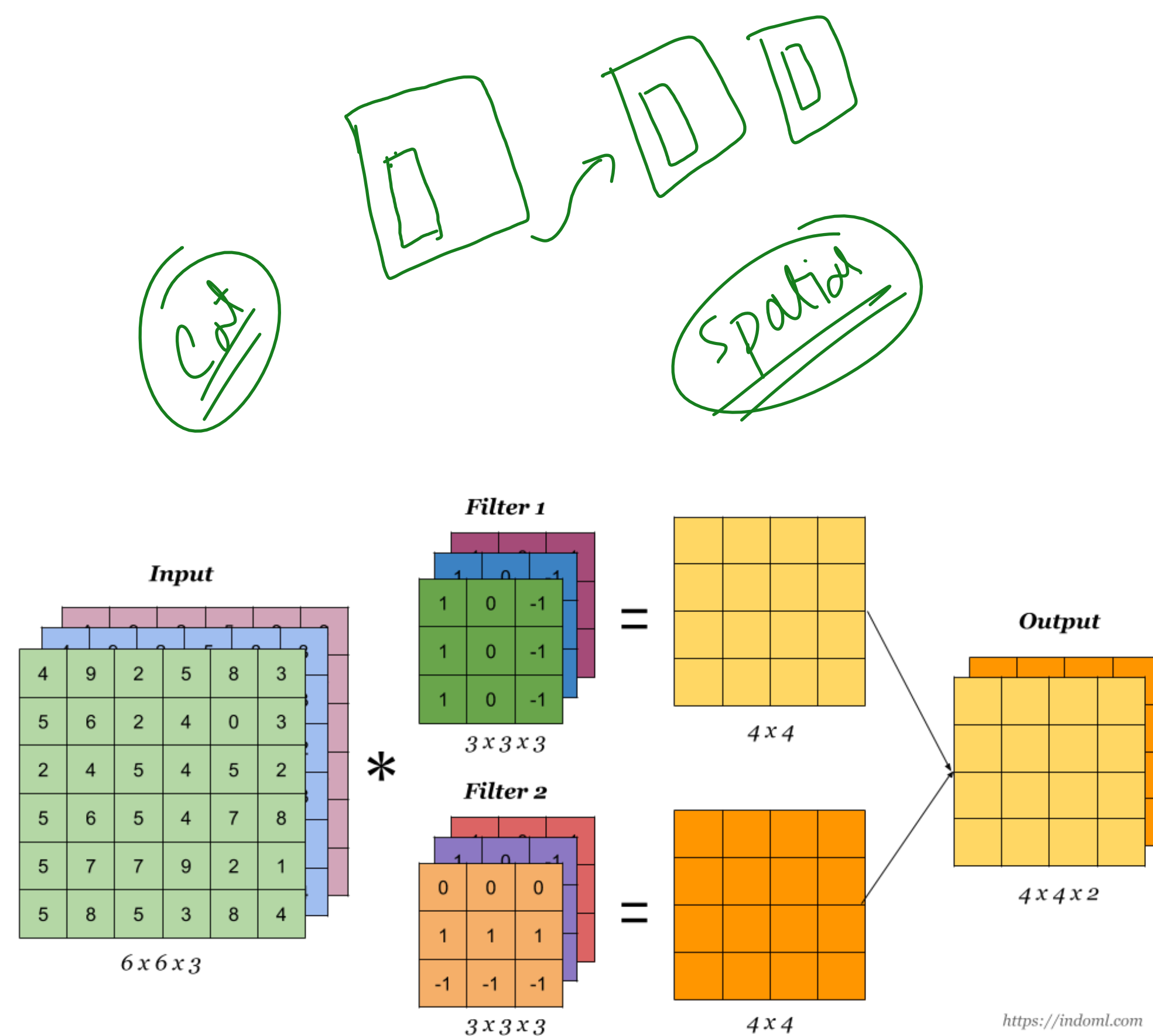


Pooling



$$228 \times 228 \times 3 \xrightarrow{\text{pooling}} 3 \times 3 \rightarrow (226 \times 226 \times 100) \times 32$$

Problem of convolution without pooling

1. Feature map size remains large
  - Convolution (with stride = 1) does not reduce size
  - Feature maps stay large across layers
  - Effect:
    - High memory usage
    - Slow computation
2. No downsampling
  - Model keeps full spatial resolution
  - Does not compress information
  - Effect:
    - Redundant information is preserved
    - Inefficient representation
3. Sensitive to small shifts
  - Small changes in input (e.g., slight movement) change output significantly
  - Effect:
    - Poor translation invariance
4. Overfitting risk increases
  - More data (large feature maps)  $\rightarrow$  more parameters in later layers
  - Effect:
    - Model may memorize instead of generalizing



5. Slower training and inference

- Large feature maps → more computations

6. Harder to capture high-level features

- Without reducing size, model focuses more on local details
- Harder to build hierarchical abstraction

Summary

Without pooling, CNNs keep large feature maps, leading to high computation, redundancy, and poor generalization.

What exactly is lost?

Example (Max Pooling 2x2):

[ 1 3  
2 4 ] → 4

- Only the **maximum value is kept**
- Other values are discarded
- So:
- Fine-grained details are lost
- Exact spatial information is reduced

Why this is acceptable

- 1. *Keeps important features*
- Max pooling keeps the **strongest activation**
- In CNNs, strong activations = important features (edges, textures)
- 2. *Reduces noise*
- Small variations are removed
- Makes the model more robust
- 3. *Provides translation invariance*
- Small shifts in input → similar pooled output
- 4. *Reduces computation*
- Smaller feature maps → faster training
- 5. *Helps generalization*
- Forces the model to focus on **important patterns**, not exact pixel values

The trade-off

| Without pooling | With pooling       |
|-----------------|--------------------|
| More detail     | Less detail        |
| More precise    | More robust        |
| Slower          | Faster             |
| Overfits more   | Generalizes better |

Important modern insight

- Yes, pooling loses information
- That's why newer architectures sometimes:
- replace pooling with **stride**
- or use **attention mechanisms**



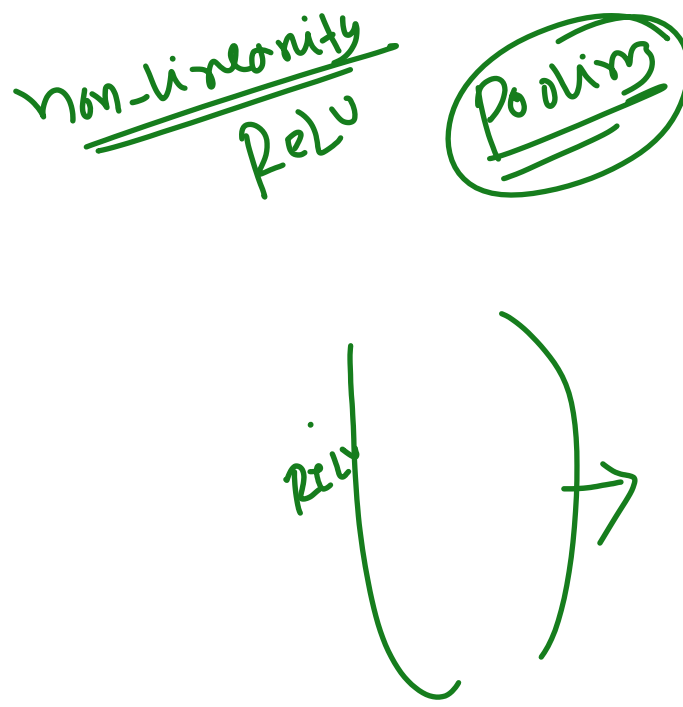


Downsampling operation means **reducing the spatial size (height and width) of an image or feature map.**

Definition

It converts a larger feature map into a smaller one:

$H \times W \rightarrow \text{smaller } H \times W$



- Convolution is linear
- ReLU makes the feature map **non-linear**
- This helps the model learn complex patterns

Types of Pooling

1. Max Pooling (most common)
- Takes the **maximum value** from each window
- Example:
- [1 3
- 2 4] → 4
- Use:
- Keeps strongest features (edges, textures)

2. Average Pooling
- Takes the **average of values**
- Example:
- [1 3
- 2 4] → (1+3+2+4)/4 = 2.5
- Use:
- Smooths features
  - Less aggressive than max pooling

3. Global Max Pooling
- Takes **maximum over entire feature map**
- Example:
- H\*W → 1 value per channel
- Use:
- Reduces each channel to a single value

4. Global Average Pooling (GAP)
- Takes **average over entire feature map**
- Use:
- Replaces fully connected layers
  - Used in modern architectures

5. Min Pooling (rare)
- Takes **minimum value**
- Use:
- Not commonly used in practice

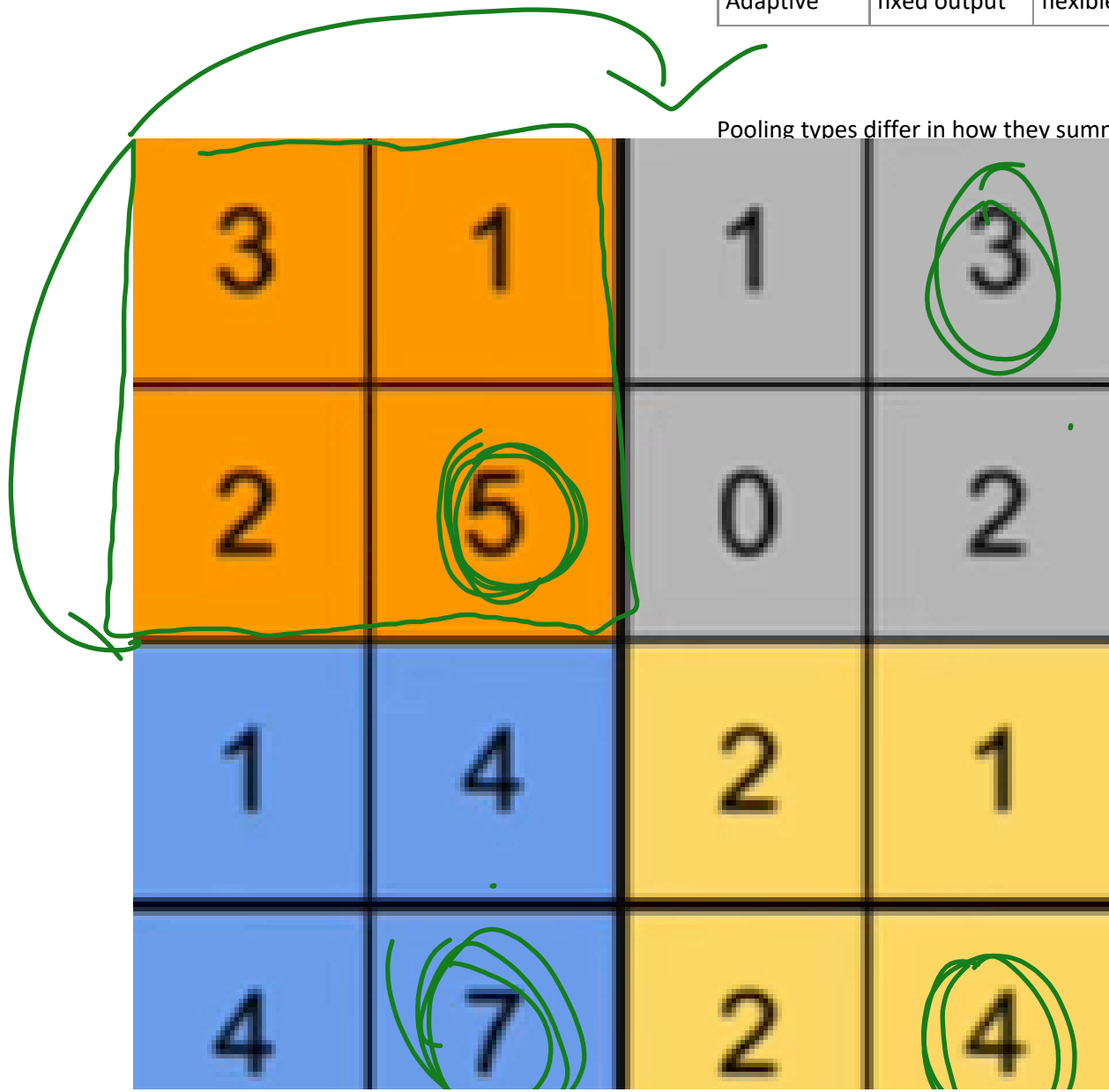
6. L2 Pooling (less common)
- Takes square root of sum of squares
- Formula:
- $\sqrt{x_1^2 + x_2^2 + \dots}$
- Use:
- Keeps energy information

7. Adaptive Pooling Output size is fixed (you define it)
- Example:
- any size → 7x7
- Use:
- Useful when input sizes vary

Summary

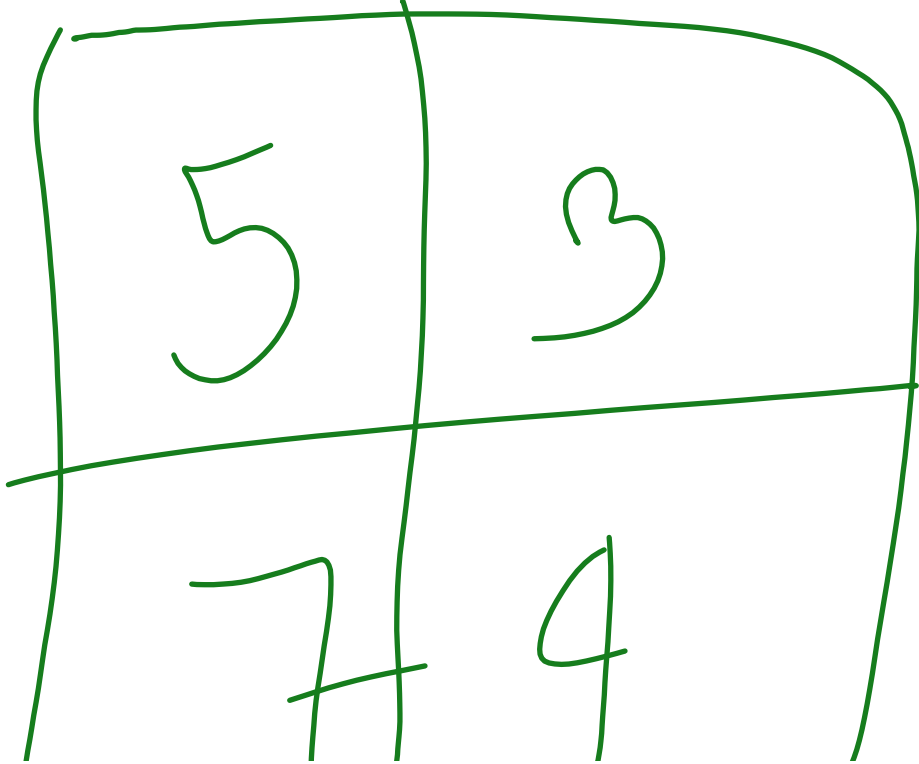
| Type        | Operation     | Usage               |
|-------------|---------------|---------------------|
| Max Pooling | max           | most common         |
| Avg Pooling | mean          | smoothing           |
| Global Max  | max over all  | feature compression |
| Global Avg  | mean over all | modern CNNs         |
| Min Pooling | min           | rare                |
| L2 Pooling  | energy        | uncommon            |
| Adaptive    | fixed output  | flexible input      |

Pooline tyoes differ in how they summarize a region, with max and average pooling being the most



size = (2,2)  
stride = 2  
type = max

max  
average







4x4

2+2

2+2

OneNote